

ROLJOHN FRILLES

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building full-stack web applications and deep learning systems. Background includes government data management platforms, an advanced neural network flood-forecasting model deployed as a cloud microservice, and mobile app development. Proficient in Python, JavaScript, PHP, and modern ML/data frameworks.

WORK EXPERIENCE

Software Developer Intern | Department of Science and Technology (DOST-CO-PES-ITD)

Taguig City, Philippines | 2025

- Developed 7 full-stack library modules (Institutions, Countries, DOST Priority Areas, Local Regions & Provinces, Areas of Expertise, Specializations, Degree Courses) with server-side data handling, validation, and interactive UIs for the Balik Scientist Program using Laravel.
- Engineered a custom WordPress plugin for the Science, Technology, and Innovation Statistics (STIS) data management system, supporting multi-step data entry for STI indicators (e.g., GRA) across multiple sectors (Government, Higher Education, Private Non-Profit, Private Industry) and reporting years.
- Built a Vue.js frontend for complex data entry, integrating Handsontable.js for spreadsheet-style manipulation and Chart.js for real-time visualization of multi-year, multi-sector STI datasets.
- Developed PHP backend functions and WordPress AJAX endpoints to securely handle data saving and retrieval, replacing a manual, spreadsheet-based reporting process with a streamlined digital workflow.

PROJECTS

HydroCast: Marikina River Basin Flood Forecasting | Mar 2025 – Dec 2025

- Engineered a hybrid deep learning architecture (STGAE, GAT, Transformer) to forecast river water levels up to 48 hours ahead using 13 years of hydrometeorological data.
- Handled 68.9% of missing real-world sensor data via unsupervised pre-training for signal imputation.
- Deployed the inference engine as a cloud-ready FastAPI/Docker microservice, decoupling ML processing from the mobile frontend.

Kuwartrack: Personal Finance Manager | Dec 2024 - Feb 2025

- Led a team of 3 building a financial tracking mobile app with a weekly expense dashboard, budget-setting transaction module, and category breakdown analytics.

Smart Logistics Optimizer (Knapsack Algorithm) | Mar 2024 – Jul 2024

- Built a mobile logistics app applying the Knapsack Algorithm to optimize truck/pallet loading for space utilization and profit margins.

EDUCATION

B.S. in Computer Science | Polytechnic University of the Philippines - Manila

Manila, Philippines | Oct 2022 - Sep 2026 (Expected)

- Completed all academic requirements; graduation formally scheduled for September 2026
- Consistent President's Lister
- DOST-SEI Scholar

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Technical Speaker, Push2Prod 2026 | DEVCON Manila Chapter

Jun 2026

- Delivered a 25-minute technical talk, "Preparing for the Real World: Testing, Improving, and Deploying Your Project," to 150+ IT/CS students, covering deployment of ML architectures into production cloud microservices.

ML/AI Co-Head | AWS Cloud Club Skill Building - PUP Manila

2023 - 2024

- Led hands-on workshops with 20+ participants on foundational Python for AI/ML, covering NumPy, Matplotlib, and Pandas.

Co-Facilitator | Collabothon: IdeaJam

Feb 2024

- Co-facilitated brainstorming sessions guiding 5+ participants through emerging-tech solution design.

SKILLS

Languages & Frameworks: Python, JavaScript, HTML/CSS, PHP, Flutter, Laravel

Data Science & ML: PyTorch, NumPy, Pandas, Matplotlib, Neural Networks, Regression, Classification, Selenium, BeautifulSoup

Backend & Deployment: FastAPI, Docker, Render, Gradio

CERTIFICATIONS & PUBLICATIONS

- Computer Vision: Deep Learning with PyTorch - Object Localization & Image Segmentation, Coursera Project Network (2023)
- "My First Exposure to Machine Learning with PyTorch," [Tutorials Dojo](#) - Technical Author (Jun 2026)